

psql Cheatsheet

// quick reference

psql cheatsheet

PostgreSQL interactive terminal
v14+ · All commands case-insensitive
Backslash cmds need no semicolon

Connecting

psql -U user -d dbname connect as user to db

psql -h host -p 5432 specify host and port

psql -U user dbname connect to postgres db

psql "postgres://u:p@h/db" connect via URI

\c dbname switch database

\c db user host switch db, user, host

\conninfo show current connection

\quit psql

Listing & Describing

\list databases

\dtlist tables

\dt schema.* list tables in schema

\dvlist views

\dilist indexes

\dslist sequences

\dflist functions

\dnlist schemas

`\dulist roles/users`
`\d tabledescribe table`
`\d+ tabledescribe with extras`
 Output & Formatting
`\xtoggle expanded output`
`\x autoauto expanded mode`
`\atoggle aligned output`
`\ttoggle tuples only`
`\timingtoggle query timing`
`\pset null 'NULL'display NULLs visibly`
`\o file.txtsend output to file`
`\oreset output to stdout`
 Query Execution & History
`\eopen query in editor`
`\ef funcnameedit function in editor`
`\ev viewnameedit view in editor`
`\gre-execute last query`
`\g file.txtexec & save to file`
`\pshow query buffer`
`\rclear query buffer`
`\sshow history`
`\s file.txtsave history to file`
 Import & Export
`\i file.sqlexecute SQL from file`
`\ir file.sqlexecute relative to current file`
`\copy t FROM 'f.csv' CSV HEADERimport CSV (client-side)`
`\copy t TO 'f.csv' CSV HEADERexport to CSV (client-side)`
`COPY t FROM '/path' CSV HEADER;server-side import (needs superuser)`
`COPY t TO '/path' CSV HEADER;server-side export`
 Variables
`\set name valset a variable`

`\setlist` all variables
`\unset name`unset a variable
`:name`use variable in SQL
`:'name'`use as quoted string
`:"name"`use as quoted ident
DDL
`CREATE DATABASE db`;create a database
`DROP DATABASE db`;drop a database
`CREATE TABLE t (...)`;create a table
`DROP TABLE t`;drop a table
`TRUNCATE t`;empty a table fast
`ALTER TABLE t ADD col`;add column
`ALTER TABLE t DROP col`;drop column
`CREATE INDEX ON t(col)`;create index
DML
`INSERT INTO t VALUES (...)`;insert row
`INSERT ... ON CONFLICT DO ...`upsert
`UPDATE t SET c=v WHERE ...`;update rows
`DELETE FROM t WHERE ...`;delete rows
`SELECT * FROM t LIMIT 10`;query rows
`EXPLAIN ANALYZE SELECT ...`;query plan + timing
Transactions
`BEGIN`;start transaction
`COMMIT`;commit transaction
`ROLLBACK`;roll back transaction
`SAVEPOINT sp`;create savepoint
`ROLLBACK TO sp`;roll back to savepoint
`RELEASE SAVEPOINT sp`;release savepoint
Roles & Permissions
`CREATE USER u WITH PASSWORD 'p'`;create user
`CREATE ROLE r`;create role

GRANT r TO u;grant role to user
 REVOKE r FROM u;revoke role
 GRANT SELECT ON t TO u;grant table privilege
 GRANT ALL ON ALL TABLES TO u;grant all on all tables
 REVOKE ALL ON t FROM u;revoke table privilege
 DROP USER u;drop user
 Useful System Queries
 SELECT version();postgres version
 SELECT current_database();current db name
 SELECT current_user;current user
 SELECT now();current timestamp
 SELECT * FROM pg_stat_activity;active connections
 SELECT * FROM pg_locks;current locks
 SELECT * FROM pg_stat_user_tables;table stats
 SELECT pg_size_pretty(pg_database_size('db'));db size
 SELECT pg_size_pretty(pg_total_relation_size('t'));table size incl. indexes
 SELECT * FROM pg_indexes WHERE tablename='t';indexes on table
 Append + to most \d commands for verbose output · Use \? for full psql
 help · Use \h CMD for SQL syntax help
 psql cheatsheet